ASSIGNMENT – VIII

TITLE :
Set up a local Kubernetes cluster with Minikube to deploy and scale your Dockerized AI application.

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admin1@bflcomp20: ~/ai-kubernetes

```
(base) admin1@bflcomp20:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
168 packages can be upgraded. Run 'apt list --upgradable' to see them.
(base) admin1@bflcomp20:~$ sudo apt install docker.io -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker.io is already the newest version (28.2.2-0ubuntu1~24.04.1).
The following packages were automatically installed and are no longer required:
  libatomic1:i386 libbsd0:i386 libdrm-amdgpu1:i386 libdrm-intel1:i386
  libdrm2:i386 libedit2:i386 libegl-mesa0:i386 libegl1:i386 libelf1t64:i386
  libexpat1:i386 libffi8:i386 libgbm1:i386 libgdk-pixbuf-xlib-2.0-0
  libgdk-pixbuf2.0-0 libgl1:i386 libgl1-mesa-dri:i386 libgles2:i386
  libglvnd0:i386 libglx-mesa0:i386 libglx0:i386 libicu74:i386 libllvm20:i386
  libnvidia-cfg1-545 libnvidia-common-545 libnvidia-extra-545
  libnvidia-fbc1-545 libnvidia-fbc1-545:i386 libopengl0:i386
  libpciaccess0:i386 libpkgconf3 libsensors5:i386 libstdc++6:i386
  libvulkan1:i386 libwayland-client0:i386 libwayland-server0:i386
  libx11-6:i386 libx11-xcb1:i386 libxau6:i386 libxcb-dri3-0:i386
  libxcb-glx0:i386 libxcb-present0:i386 libxcb-randr0:i386 libxcb-shm0:i386
  libxcb-sync1:i386 libxcb-xfixes0:i386 libxcb1:i386 libxdmcp6:i386
  libxext6:i386 libxml2:i386 libxshmfence1:i386 libxxf86vm1:i386 mailcap
  mesa-libgallium:i386 mesa-vulkan-drivers:i386 nvidia-modprobe nvidia-prime
  nvidia-settings pkg-config pkgconf pkgconf-bin screen-resolution-extra
  xserver-xorg-video-nvidia-545
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 168 not upgraded.
(base) admin1@bflcomp20:~$ sudo systemctl start docker
(base) admin1@bflcomp20:~$ sudo systemctl enable docker
(base) admin1@bflcomp20:~$ sudo usermod -aG docker $USER
(base) admin1@bflcomp20:~$ newgrp docker
(base) admin1@bflcomp20:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
(base) admin1@bflcomp20:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
(base) admin1@bflcomp20:~$ sudo snap install kubectl --classic
kubectl 1.34.5 from Canonical** installed
```

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```
(base) admin1@bflcomp20:~$ kubectl version --client
Client Version: v1.34.5
Kustomize Version: v5.7.1
(base) admin1@bflcomp20:~$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  128M  100  128M    0     0  11.9M      0  0:00:10  0:00:10 --:--:-- 15.1M
(base) admin1@bflcomp20:~$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
(base) admin1@bflcomp20:~$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
(base) admin1@bflcomp20:~$ minikube start
😄  minikube v1.38.1 on Ubuntu 24.04
✨  Automatically selected the docker driver. Other choices: ssh, none
❗  Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
📌  Using Docker driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
💾  Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 15.00 M
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 19.10 M
🔥  Creating docker container (CPUs=2, Memory=3900MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
(base) admin1@bflcomp20:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   17s   v1.35.1
(base) admin1@bflcomp20:~$ mkdir ai-kubernetes
(base) admin1@bflcomp20:~$ cd ai-kubernetes
(base) admin1@bflcomp20:~/ai-kubernetes$ nano app.py
```

admin1@bflcomp20: ~/ai-kubernetes

```
  GNU nano 7.2                                                    app.py *
from flask import Flask, request, jsonify

app = Flask(__name__)

@app.route('/')
def home():
    return "AI Model Running Successfully"

@app.route('/predict', methods=['POST'])
def predict():
    data = request.json['number']
    result = data * 2
    return jsonify({'result': result})

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
(base) admin1@bflcomp20:~/ai-kubernetes$ nano Dockerfile
```

admin1@bflcomp20: ~/ai-kubernetes

```
  GNU nano 7.2                                                    Dockerfile *
FROM python:3.10

WORKDIR /app

COPY . .

RUN pip install flask

EXPOSE 5000

CMD ["python","app.py"]
```

admin1@bflcomp20: ~/ai-kubernetes

```
(base) admin1@bflcomp20:~/ai-kubernetes$ eval $(minikube docker-env)
(base) admin1@bflcomp20:~/ai-kubernetes$ docker build -t ai-model .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/6 : FROM python:3.10
3.10: Pulling from library/python
866771c43bf5: Pulling fs layer
ed881fbf1b07: Pulling fs layer
9da421ddeb65: Pulling fs layer
a793e3c6bce8: Pulling fs layer
f378069ec6df: Pulling fs layer
667230d3ef7e: Pulling fs layer
2d26a4ad9473: Pulling fs layer
f378069ec6df: Waiting
a793e3c6bce8: Waiting
667230d3ef7e: Waiting
2d26a4ad9473: Waiting
ed881fbf1b07: Verifying Checksum
ed881fbf1b07: Download complete
866771c43bf5: Verifying Checksum
866771c43bf5: Download complete
866771c43bf5: Pull complete
9da421ddeb65: Verifying Checksum
9da421ddeb65: Download complete
ed881fbf1b07: Pull complete
9da421ddeb65: Pull complete
f378069ec6df: Verifying Checksum
f378069ec6df: Download complete
667230d3ef7e: Verifying Checksum
667230d3ef7e: Download complete
2d26a4ad9473: Verifying Checksum
2d26a4ad9473: Download complete
a793e3c6bce8: Download complete
a793e3c6bce8: Pull complete
f378069ec6df: Pull complete
667230d3ef7e: Pull complete
2d26a4ad9473: Pull complete
Digest: sha256:c6b2a9f40fbee6dc2ed9c2355655858ce3725c4d07aa546a102fa9ebb2d3164d
Status: Downloaded newer image for python:3.10
 ---> 10a104bae12d
Step 2/6 : WORKDIR /app
 ---> Running in b3f01d4c9285
```

admin1@bflcomp20: ~/ai-kubernetes

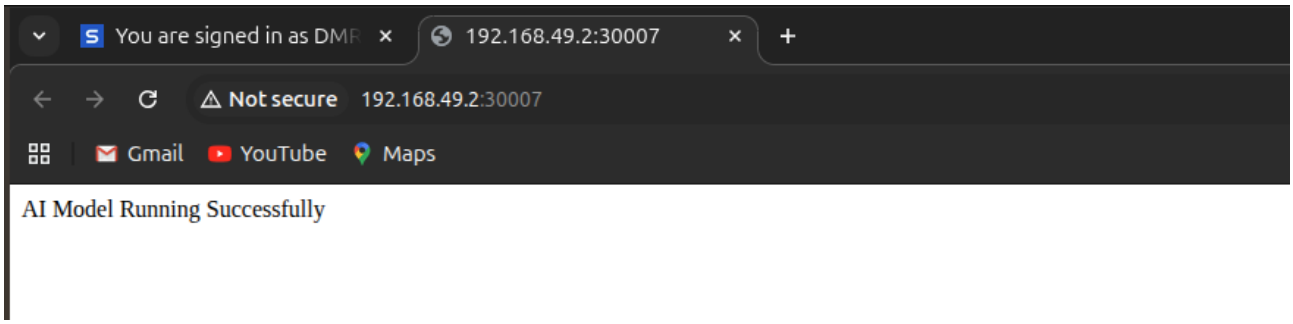
```
(base) admin1@bflcomp20:~/ai-kubernetes$ docker images
REPOSITORY                                TAG       IMAGE ID       CREATED          SIZE
ai-model                                  latest    d68748c2d443   11 seconds ago   1.11GB
python                                    3.10      10a104bae12d   7 days ago       1.09GB
registry.k8s.io/kube-apiserver            v1.35.1   6f9eeb0cff98   4 weeks ago      89.8MB
registry.k8s.io/kube-scheduler            v1.35.1   5f2a969bc7a4   4 weeks ago      51.7MB
registry.k8s.io/kube-proxy                v1.35.1   6521110cdb01   4 weeks ago      70.4MB
registry.k8s.io/kube-controller-manager   v1.35.1   8d7002962c48   4 weeks ago      75.8MB
registry.k8s.io/etcd                      3.6.6-0   0a108f71895б   3 months ago     62.5MB
registry.k8s.io/coredns/coredns           v1.13.1   aa5e3ebc0dfe   5 months ago     78.1MB
registry.k8s.io/pause                     3.10.1    cd073f4c5f6a   8 months ago     736kB
gcr.io/k8s-minikube/storage-provisioner   v5        6e38f40d628d   4 years ago      31.5MB
(base) admin1@bflcomp20:~/ai-kubernetes$ nano deployment.yaml
```

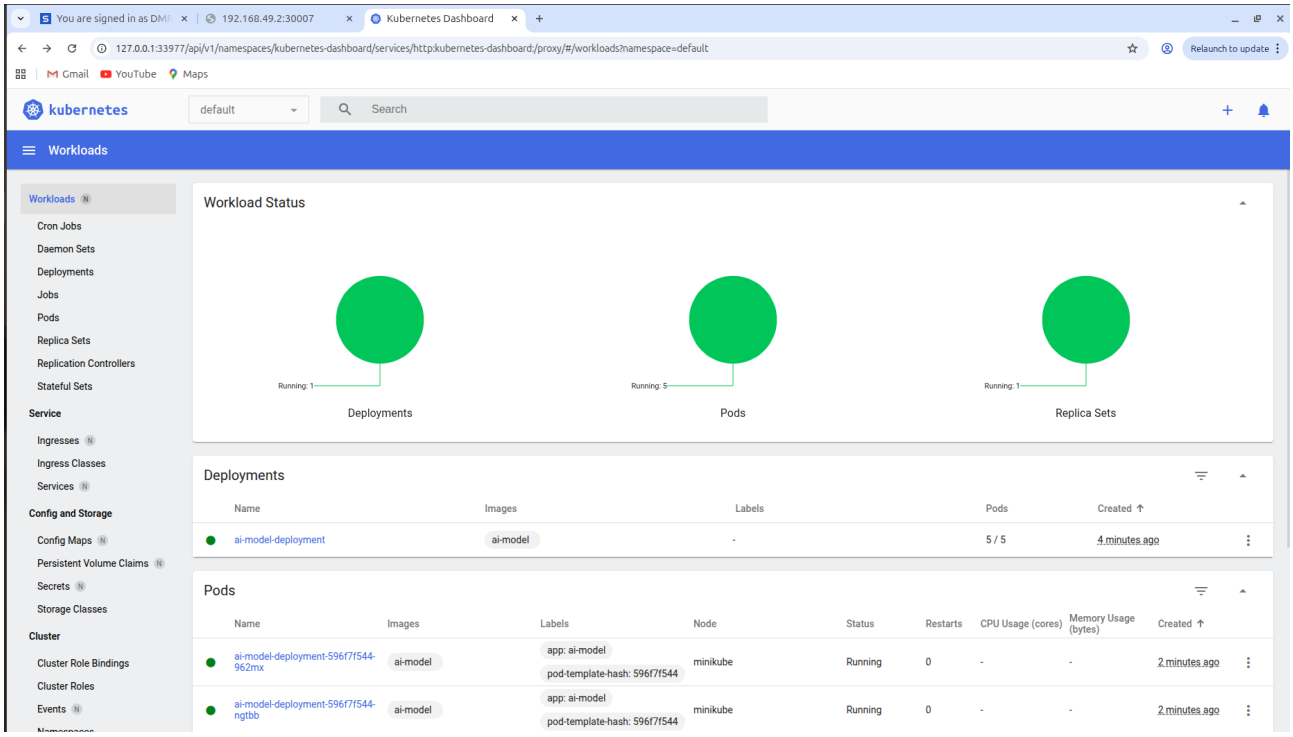
```
admin1@bflcomp20: ~/ai-kubernetes
GNU nano 7.2                                                    deployment.yaml *
apiVersion: apps/v1
kind: Deployment

metadata:
  name: ai-model-deployment

spec:
  replicas: 2

  selector:
    matchLabels:
      app: ai-model

  template:
    metadata:
      labels:
        app: ai-model

    spec:
      containers:
      - name: ai-container
        image: ai-model
        imagePullPolicy: Never
        ports:
        - containerPort: 5000
```

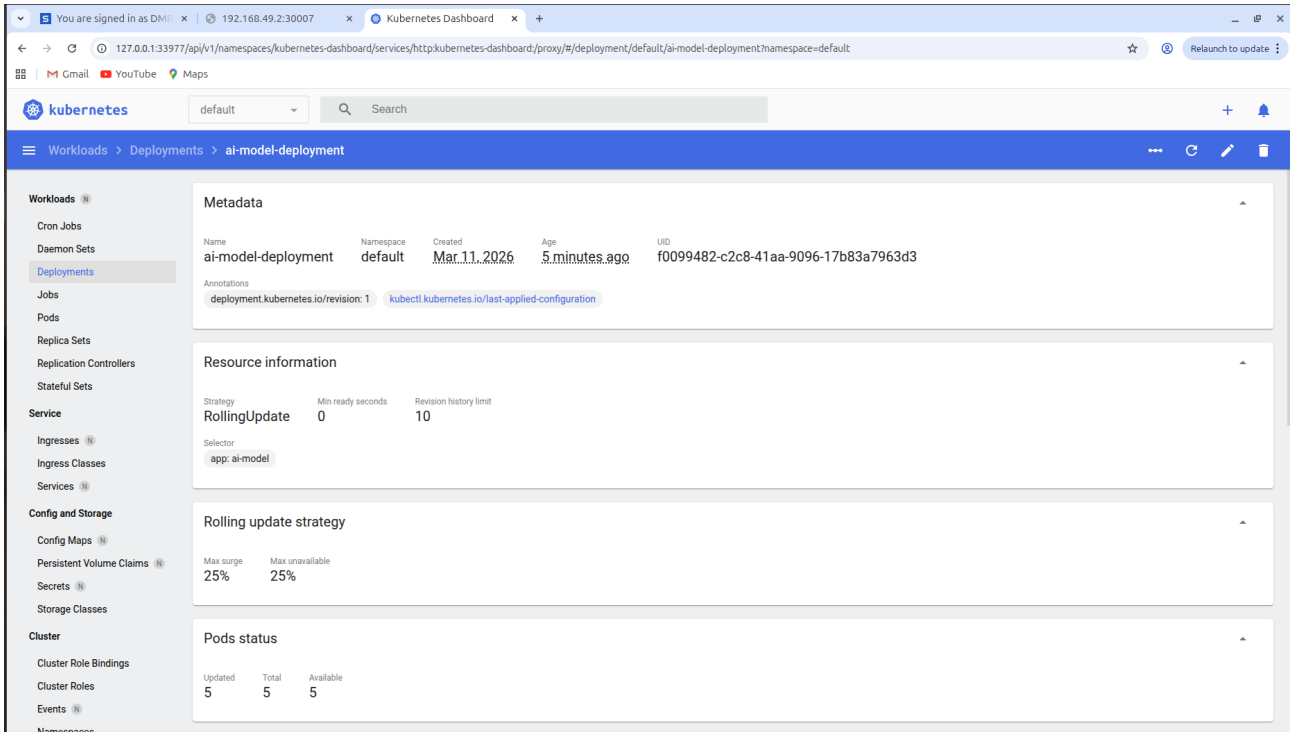
```
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/ai-model-deployment created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
ai-model-deployment-596f7f544-cbd9v    1/1     Running   0          8s
ai-model-deployment-596f7f544-wnbq2    1/1     Running   0          8s
(base) admin1@bflcomp20:~/ai-kubernetes$ nano service.yaml
```

```
admin1@bflcomp20: ~/ai-kubernetes
GNU nano 7.2                                                    service.yaml *
apiVersion: v1
kind: Service

metadata:
  name: ai-model-service

spec:
  type: NodePort

  selector:
    app: ai-model

  ports:
  - port: 5000
    targetPort: 5000
    nodePort: 30007
```

```
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f service.yaml
service/ai-model-service created
(base) admin1@bflcomp20:~/ai-kubernetes$ minikube service ai-model-service
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | ai-model-service | 5000        | http://192.168.49.2:30007 |
|-----------|-----------------|-------------|---------------------------|

🎉  Opening service default/ai-model-service in default browser...
(base) admin1@bflcomp20:~/ai-kubernetes$ Opening in existing browser session.
kubectl scale deployment ai-model-deploym kubectl scale deployment ai-model-deployment --replicas=5
deployment.apps/ai-model-deployment scaled
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
ai-model-deployment-596f7f544-962mx    1/1     Running   0          42s
ai-model-deployment-596f7f544-cbd9v    1/1     Running   0          3m33s
ai-model-deployment-596f7f544-ngtbb    1/1     Running   0          42s
ai-model-deployment-596f7f544-rw64l    1/1     Running   0          42s
ai-model-deployment-596f7f544-wnbq2    1/1     Running   0          3m33s
(base) admin1@bflcomp20:~/ai-kubernetes$ minikube dashboard
🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:33977/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Opening in existing browser session.
```


You are signed in as DMR
192.168.49.2:30007
Not secure
192.168.49.2:30007
Gmail
YouTube
Maps
AI Model Running Successfully


Kubernetes Dashboard
127.0.0.1:33977/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/workloads?namespace=default
kubernetes
default
Search
Workloads
Workload Status
Deployments
Pods
Replica Sets
Deployments
Name
Images
Labels
Pods
Created
ai-model-deployment
ai-model
5 / 5
4 minutes ago
Pods
ai-model-deployment-596f7f544-962mx
ai-model-deployment-596f7f544-ngtbb
app: ai-model
pod-template-hash: 596f7f544
minikube
Running
2 minutes ago


Kubernetes Dashboard
127.0.0.1:33977/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/deployment/default/ai-model-deployment?namespace=default
kubernetes
default
Search
Workloads > Deployments > ai-model-deployment
Metadata
Name
ai-model-deployment
Namespace
default
Created
Mar 11, 2026
Age
5 minutes ago
UID
f0099482-c2c8-41aa-9096-17b83a7963d3
Annotations
deployment.kubernetes.io/revision: 1
kubectl.kubernetes.io/last-applied-configuration
Resource information
Strategy
RollingUpdate
Min ready seconds
0
Revision history limit
10
Selector
app: ai-model
Rolling update strategy
Max surge
25%
Max unavailable
25%
Pods status
Updated
5
Total
5
Available
5